

Wednesday

Tools & Practice

Nash → Selten → Harsanyi

Instructor: Prof. Luyao Zhang · Repository 1: [sunshineluyao/gt-tools-demos](https://github.com/sunshineluyao/gt-tools-demos)

Two notebooks, one learning route. Use QuantEcon + Nashpy for payoff matrices, then Gambit/PyGambit for credible continuation and private costs. Predict, compute, change one assumption, explain.

Practice block · suggested 75 minutes

Min.	Activity	Student evidence
0-5	Choose the representation	Identify timing, payoff knowledge and hidden information.
5-20	Notebook 01 · QuantEcon + Nashpy	Verify prisoner's dilemma and matching pennies; edit one payoff.
20-35	Notebook 02 · Selten entry	Compare pure Nash profiles with backward-induction SPNE.
35-50	Notebook 02 · Harsanyi types	Inspect information sets; change the prior and costs.
50-65	Gambit desktop + GTE introduction	Open the exported files and inspect the graphical representations.
65-75	Interdisciplinary exit ticket	Record a changed input, prior prediction, output and explanation.

Choose the model before the solver

Game	Concept	This week
Static + complete information	Nash equilibrium	Notebook 01: payoff matrix
Dynamic + complete information	Subgame-perfect Nash equilibrium	Notebook 02: entry and backward induction
Static + incomplete information	Bayesian Nash equilibrium	Notebook 02: private-cost entry
Dynamic + incomplete information	Beliefs and further refinements	Future topic; no additional demo this week

Before class: run both setup cells. PyGambit may compile from source on Linux and take several minutes; compilation is outside the 75-minute activity estimate. Use CPU. No API key or GPU is required.

01 · Matrix incentives

02 · Credible continuation

Simple, fully annotated code with editable game parameters.

Nash · one matrix, two libraries

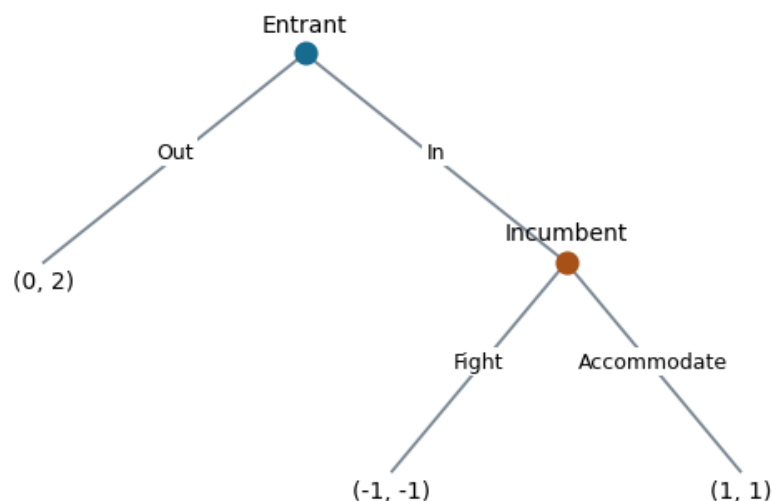
Row / Column	Cooperate	Defect
Cooperate	(3, 3)	(0, 5)
Defect	(5, 0)	(1, 1)

The first payoff belongs to the row player. Both players defect, receiving (1,1). For matching pennies, each randomizes equally and receives expected payoff 0. Notebook 01 checks candidates from both libraries against unilateral deviations.

Try: select Coordination, edit any of the eight payoff boxes, then click **Solve and check**. QuantEcon stores each player's own action first, so its second player receives B.T; Nashpy receives A and B directly.

Selten · a threat must be credible

Selten: entry and credible continuation
Dashed line = nodes the acting player cannot distinguish



Pure Nash: (Out, Fight) and (In, Accommodate).

Unique baseline SPNE: (In, Accommodate), because $1 > -1$ for the incumbent after entry.

Change the game: set the incumbent's Fight payoff to 2. Fighting becomes credible and the entrant stays Out. At ties the code keeps every pure best response. The Gambit Nash solver and the separate backward-induction check answer different questions.

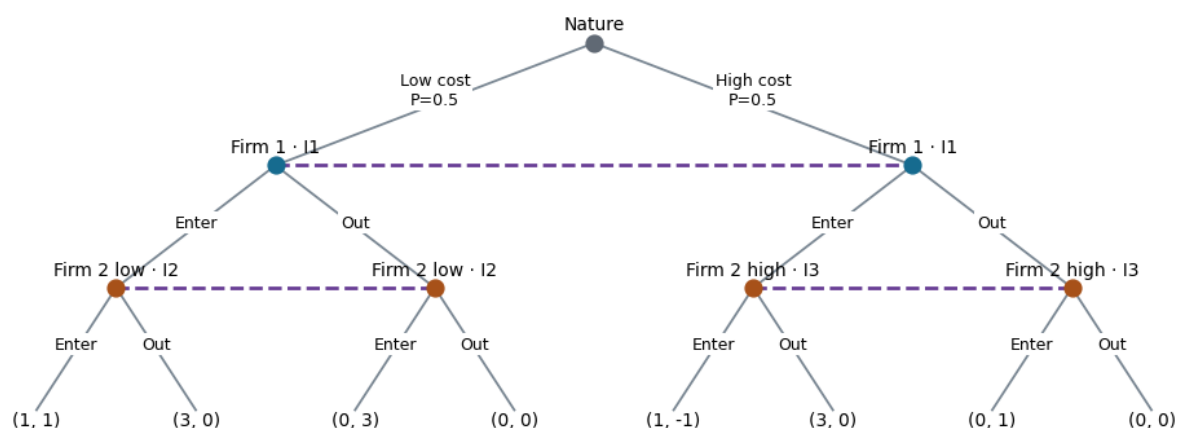
Sources: Nash (1950), [PNAS](#); Selten (1965), [original article](#). Numerical examples follow the class slides.

02 · Private information

Harsanyi: types, a common prior and contingent plans.

Firm 1's cost is 1. Firm 2's cost is 1 or 3, each with probability 1/2. Revenue is 4 for a sole entrant and 2 for each firm if both enter. Entry incurs cost; staying Out pays 0.

Harsanyi: hidden cost and simultaneous entry
Dashed line = nodes the acting player cannot distinguish



Read the dashed lines: Firm 1 cannot observe Firm 2's cost. Firm 2 knows its own cost but cannot observe Firm 1's entry choice. Its two type-specific information sets encode those assumptions.

Baseline Bayesian equilibrium: Firm 1 enters; Firm 2 enters if low cost and stays Out if high cost. Firm 1's expected payoff is $0.5 \times (2 - 1) + 0.5 \times (4 - 1) = 2$. Firm 2's ex-ante expected payoff is **0.5**.

Explore, check, explain

1. Change $P(\text{low cost})$ to 0.8, keeping costs fixed. The same pure actions remain optimal; Firm 1's expected payoff falls to 1.4.
2. Change Firm 1's cost or the high type's cost. Check each type's incentive separately.
3. Read the result as a complete contingent plan. The notebook enumerates pure equilibria; an empty list does not rule out mixed equilibria.

Take the model into Gambit

Open **examples/bayesian_baseline.efg** in the desktop app. Find Nature and all three strategic information sets. Change parameters in the notebook, click **Rebuild and solve**, then download and reopen **game_exports/bayesian_current.efg**.

Representation reminder: this is a simultaneous Bayesian game drawn as a tree. Drawing an action earlier does not make it observable. A generic Nash computation is not automatically a dynamic-game refinement.

Source: Harsanyi (1967), [Games with incomplete information, Part I](#). Original classroom visualization generated from the executable game.

Graphical tools & classroom checklist

Gambit desktop · 10-minute walkthrough

Open **examples/entry_baseline.efg**. Identify players, actions and terminal payoffs. Choose **Tools → Equilibrium**, then inspect **View → Profiles**. Compare the pure profiles with the notebook's SPNE answer. Open the Bayesian file and inspect information sets; **Format → Labels** controls labels.

The desktop application is installed separately. Installing **pygambit** supplies the Python interface, not the desktop UI. Menu references follow the official 16.7.0 documentation; the desktop GUI itself was not launched during preparation.

Game Theory Explorer · introduce and preflight

GTE is a graphical web tool for building and analyzing strategic-form matrices and extensive-form trees, with Nash equilibrium algorithms. It is a separate project, licensed **GPL-3.0**.

[GTE landing page](#) · [Direct web builder](#) · [Source repository](#)

The source repository documents a legacy Flash GUI. Live builder operation was not verified. Check the URLs before class; if the builder does not render, continue with the same models in Gambit. No Flash installation is included.

Required citation: Rahul Savani and Bernhard von Stengel (2015), Game Theory Explorer – Software for the Applied Game Theorist, Computational Management Science 12, 5–33. [doi:10.1007/s10287-014-0206-x](https://doi.org/10.1007/s10287-014-0206-x).

Tool	License	Software citation
QuantEcon.py 0.11.4	MIT	Batista et al. (2024), JOSS 9(93), 5585.
Nashpy 0.0.43	MIT	Knight & Campbell (2018), JOSS 3(30), 904.
Gambit/PyGambit 16.7.0	GPL-2.0-or-later	Savani & Turocy (2026), Gambit package.
GTE	GPL-3.0	Savani & von Stengel (2015), above.

Exit ticket · three disciplinary lenses

Economics: which incentive or welfare comparison changed?

Computer science: how did the representation and algorithm shape the result?

Behavioral science: what observable choice might depart from the benchmark?

Future folders stay empty: OpenSpiel, Axelrod, Mesa, PettingZoo, RLlib and oTree. The repository README contains their roles, license links, the software-selection decision tree and full references, including actual Nashpy and Gambit use in research papers.

Validation: both notebooks ran locally with saved outputs; 17 model/presentation tests passed, including 27 Bayesian parameter combinations. Remote Colab rendering and student-account access remain untested.